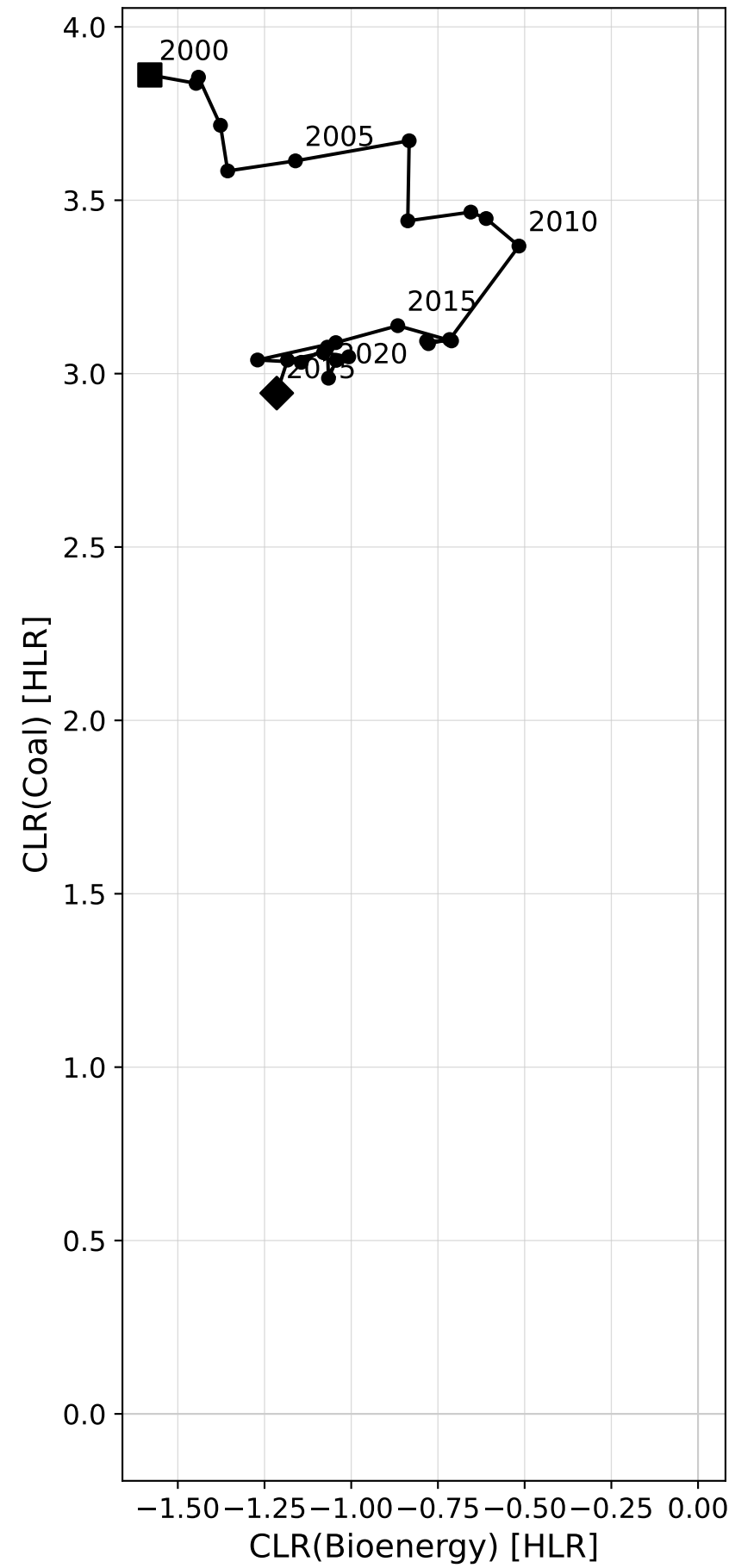


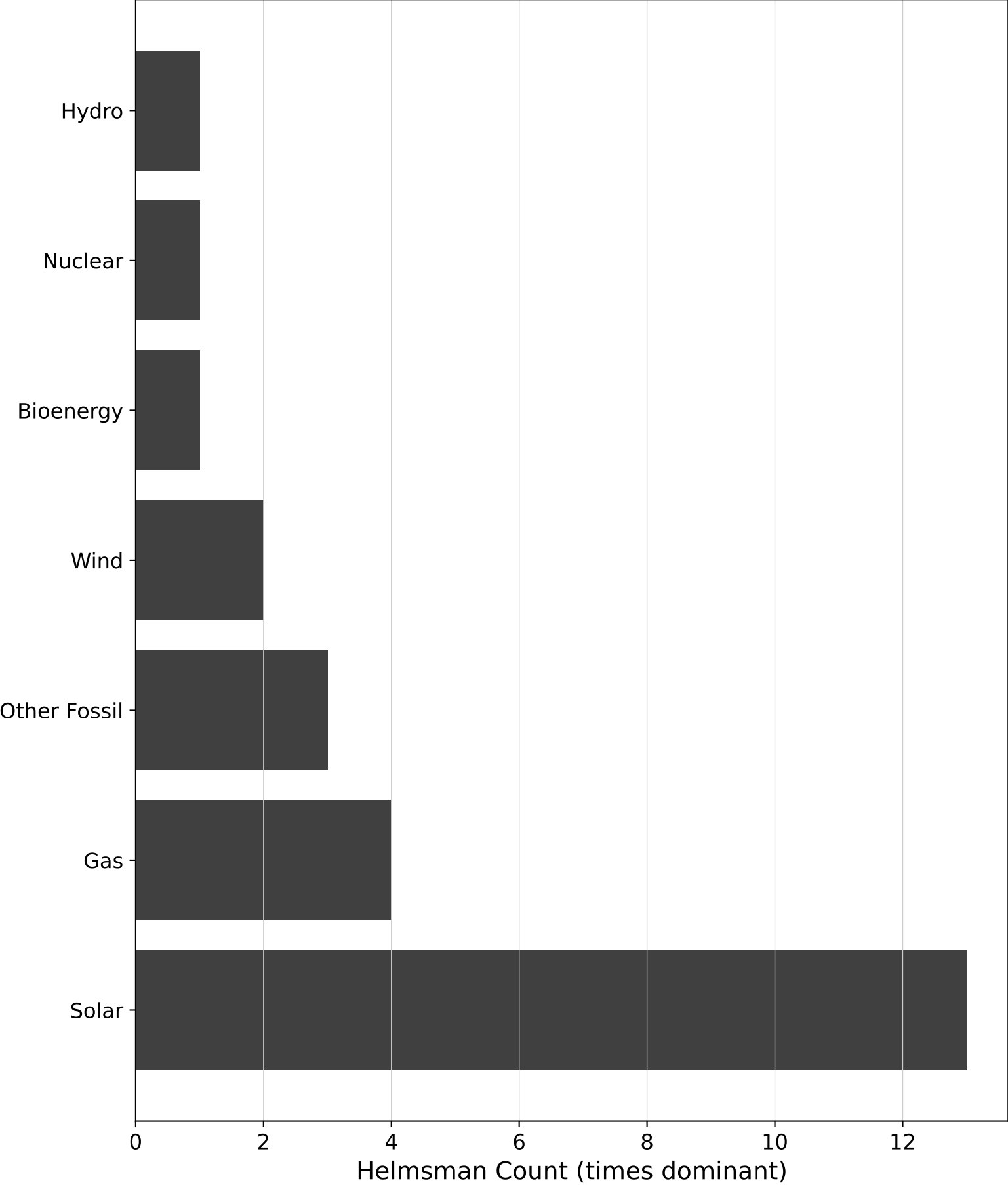
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



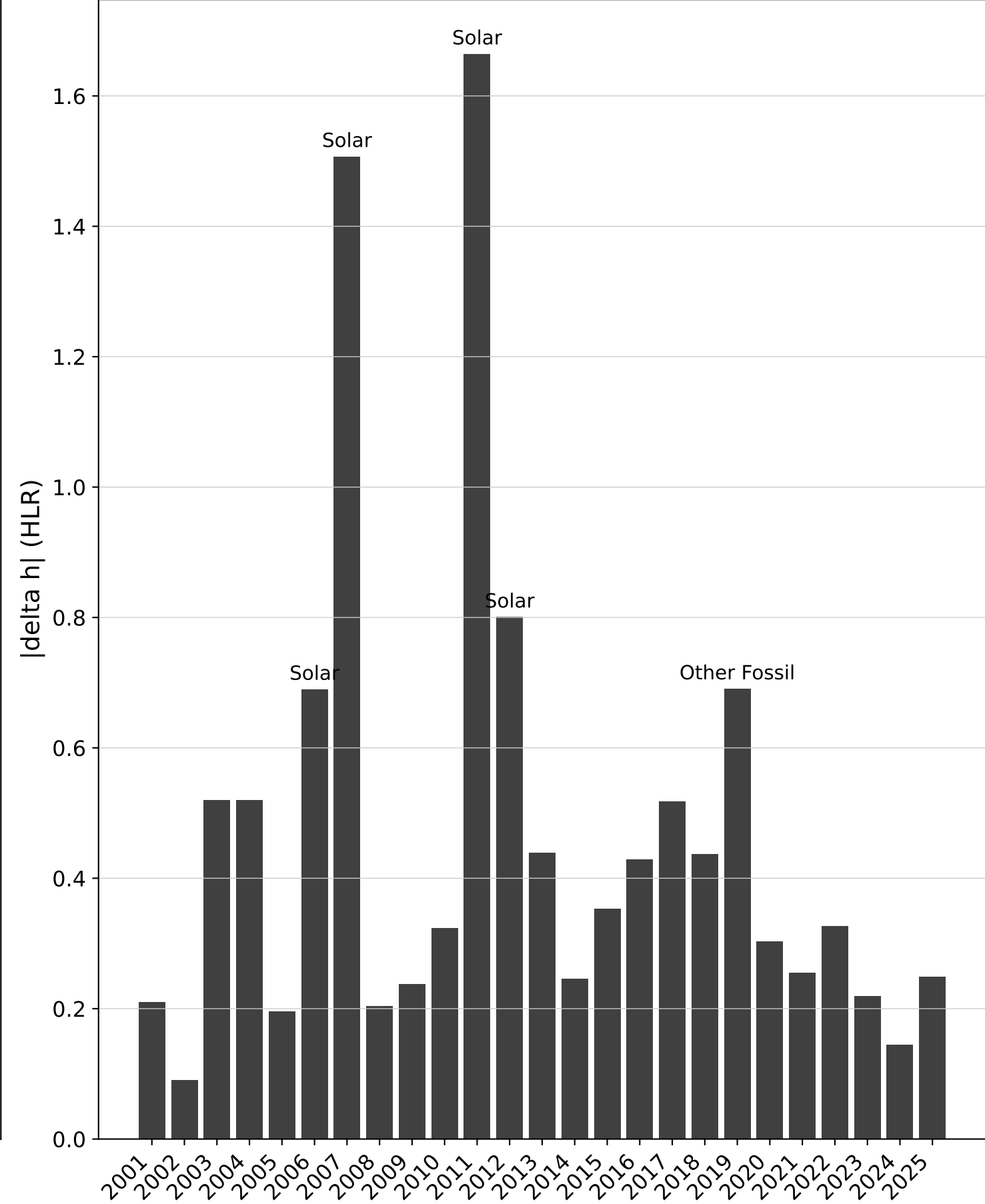
STAGE 2 – Aitchison Distance Matrix (all year pairs)


$$d_A(\text{HLR})$$

HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



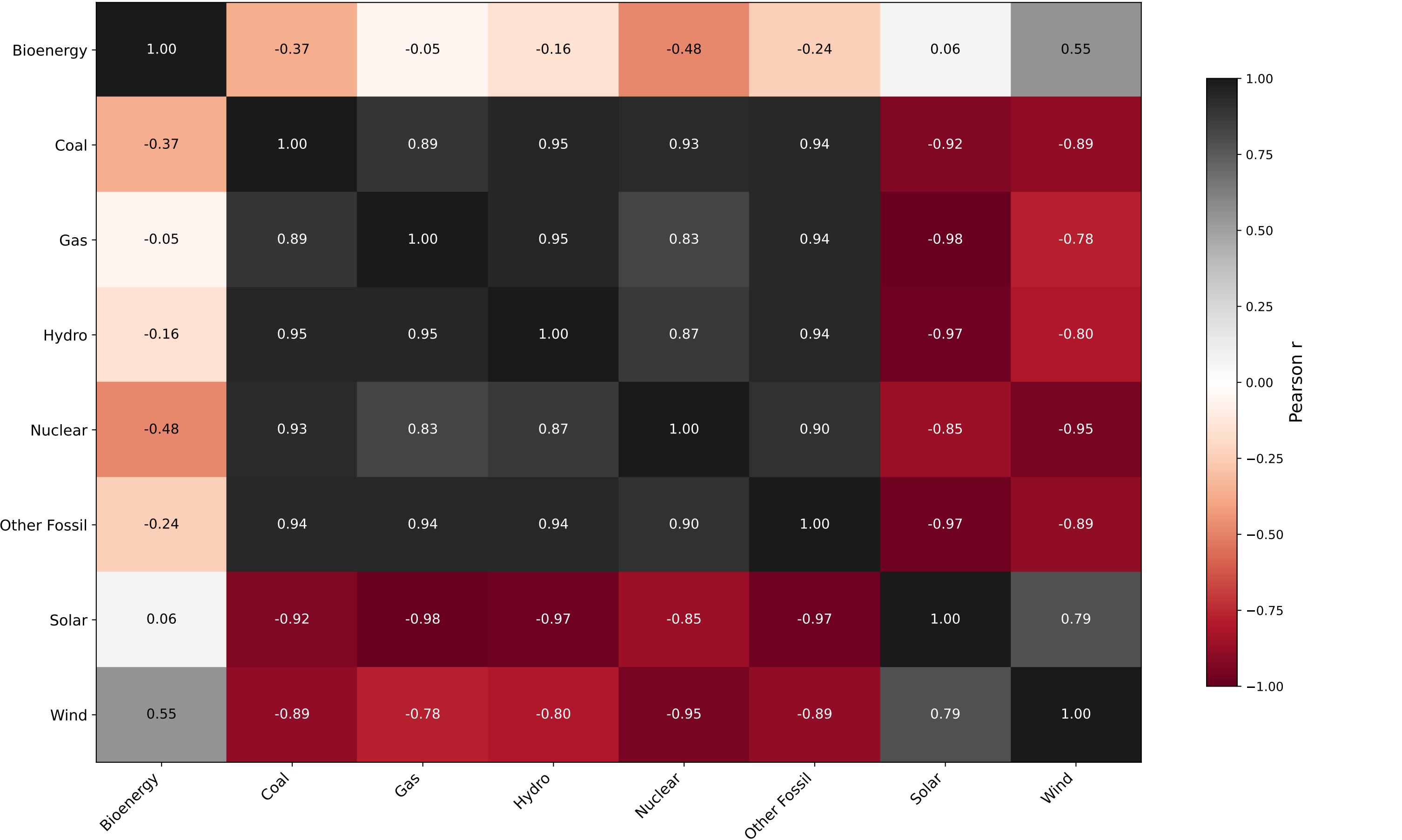
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2025	9.4603	84.8	Solar
2	2002	2025	9.4248	84.9	Solar
3	2001	2025	9.3775	84.6	Solar
4	2006	2025	9.2075	83.0	Solar
5	2000	2024	9.1358	80.8	Solar
6	2002	2024	9.0976	80.8	Solar
7	2001	2024	9.0528	80.6	Solar
8	2000	2023	9.0441	79.6	Solar
9	2002	2023	9.0023	79.6	Solar
10	2001	2023	8.9569	79.4	Solar
11	2000	2022	8.9232	77.9	Solar
12	2006	2024	8.8998	79.1	Solar
13	2003	2025	8.8960	84.2	Solar
14	2002	2022	8.8848	77.9	Solar
15	2001	2022	8.8372	77.6	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2010	2025	8.8156	2.92
2	2000	2006	2025	8.5653	5.08
3	2000	2010	2024	8.4875	2.82
4	2000	2009	2025	8.3089	3.24
5	2000	2006	2024	8.2823	4.91
6	2000	2010	2023	8.2104	2.80
7	2000	2006	2023	8.1635	4.86
8	2000	2010	2022	8.0866	2.76
9	2000	2006	2022	8.0472	4.79
10	2000	2009	2024	8.0278	3.13

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4986	Hs-3	75.9	0.00	0.0000	---	57139
2001	0.5068	Hs-4	74.5	2.30	0.3563	Wind	58609
2002	0.5095	Hs-4	75.1	0.80	0.1264	Gas	60964
2003	0.5008	Hs-4	65.8	1.84	0.6194	Solar	31906
2004	0.4746	Hs-3	65.6	4.74	0.6706	Wind	34916
2005	0.4910	Hs-3	64.6	2.26	0.3241	Bioenergy	35226
2006	0.4933	Hs-3	73.6	5.33	0.9426	Solar	74443
2007	0.4889	Hs-3	51.3	6.80	1.6922	Solar	13271
2008	0.5034	Hs-4	50.9	2.90	0.3633	Nuclear	13807
2009	0.5059	Hs-4	48.4	2.67	0.3764	Other Fossil	10983
2010	0.4933	Hs-3	45.6	3.39	0.4518	Other Fossil	8513
2011	0.4808	Hs-3	28.7	11.11	1.8198	Solar	1244
2012	0.5166	Hs-4	22.2	7.39	0.9115	Solar	518
2013	0.5385	Hs-4	19.6	6.84	0.6175	Gas	333
2014	0.5499	Hs-4	18.2	3.52	0.3124	Solar	256
2015	0.5660	Hs-4	17.8	6.28	0.4664	Solar	185
2016	0.5622	Hs-4	16.4	7.17	0.5433	Solar	148
2017	0.5504	Hs-4	16.0	8.64	0.6091	Solar	166
2018	0.5467	Hs-4	16.5	8.81	0.6227	Solar	246
2019	0.5340	Hs-4	20.4	8.09	0.7555	Other Fossil	521
2020	0.5094	Hs-4	19.9	4.23	0.3364	Solar	506
2021	0.5282	Hs-4	19.7	4.35	0.3379	Gas	495
2022	0.5270	Hs-4	20.2	5.91	0.4633	Gas	519
2023	0.5334	Hs-4	19.9	3.64	0.2859	Hydro	498
2024	0.5239	Hs-4	20.1	2.64	0.2077	Solar	498
2025	0.4849	Hs-3	20.8	5.07	0.4061	Solar	508